GURU NANAK INSTITUTE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING

"IoT BASED FOOD COLD STROAGE MONITORING AND CONTROLLING SYSTEM"

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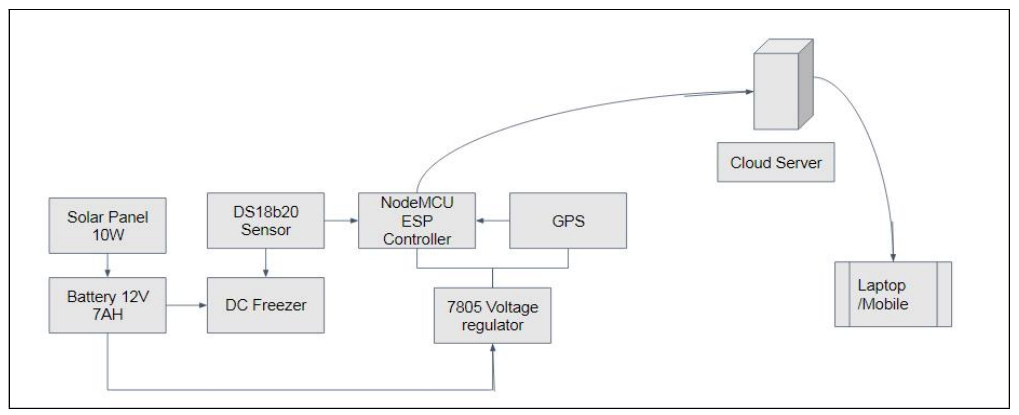
Abstract:Tricholoma matsutake (T. matsutake) is a special type of fungus known as "the king of bacteria", and has the very high economic value. However, it is also very difficult to transport due to its corruptibility. Therefore, tracing and tracking the quality and safety of T. matsutake in the cold chain is very important and necessary.Based on changes in the cold chain environmental parameters determine the safety of T. matsutake is a viable option.

Introduction: In recent years, the Internet of Things (IoT) has become an important research topic [7–11], and the emerging IoT technologies are gradually ripening. After nearly a decade of development, the definition of the IoT has become more extensive, and the application of IoT involves more fields that include supply chain management, health care, utilities, transportation and other field. In general, the IoT includes two important features: one is to collect data and connect itself to the cloud's perceived

System Block Diagram:



Database model/Implementation/USPs of the project etc.

The first step is to conduct a feasibility study and determine the requirements of the system. This includes identifying the sensors and IoT devices needed, the communication network, and the software required to collect and analyze the data

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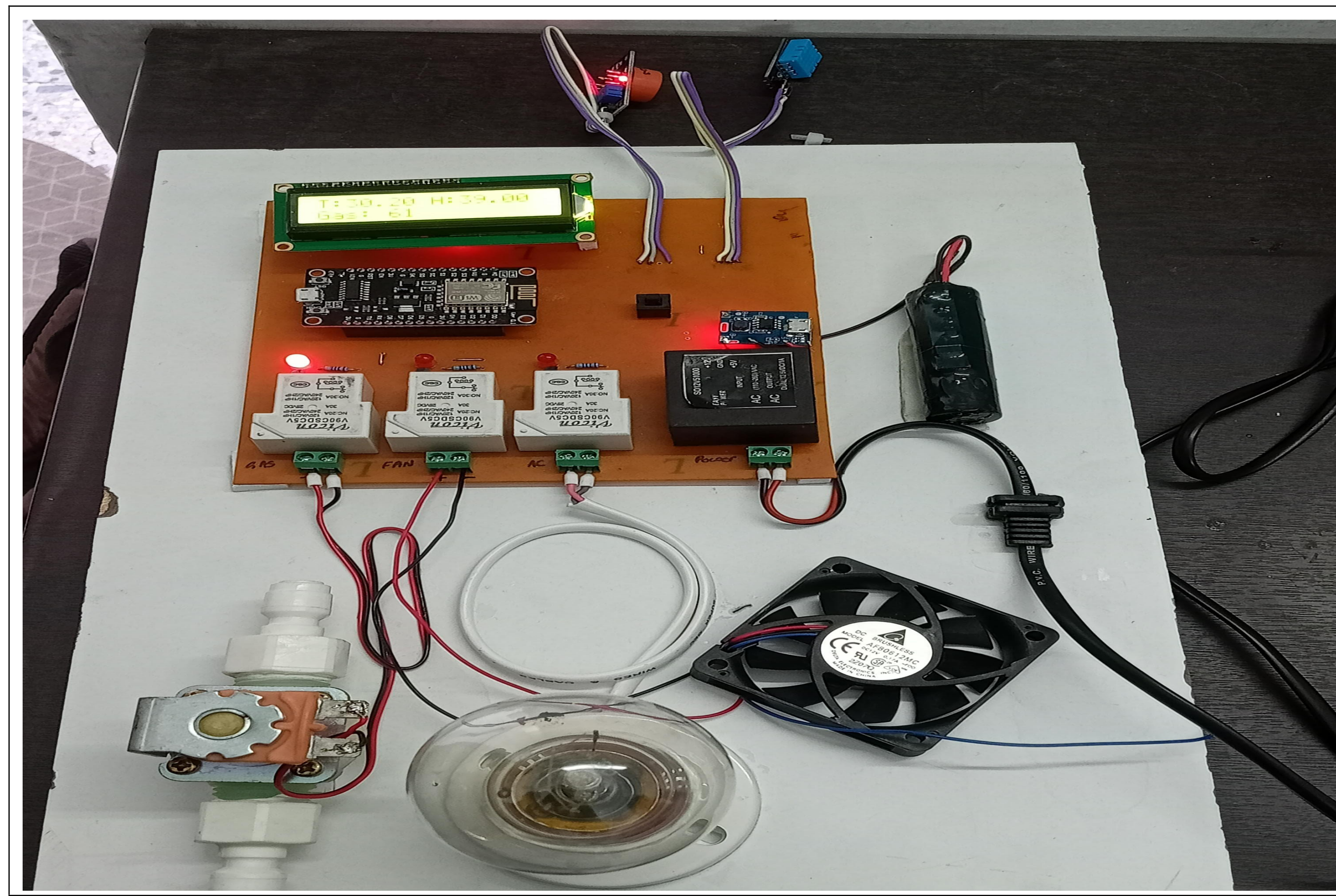
The next step is to install the hardware components of the system, including the sensors, actuators, and microcontrollers. The sensors should be placed strategically throughout the cold storage facility to ensure accurate reading

Hardware/Screenshot



Result:

As a result, the effort required to supervise manual employees can be significantly decreased, thus enhancing and guaranteeing occupational health and safety. Traditional methods, except for temperature and relative humidity, do not provide real-time monitoring of various environmental data.

a.

Conclusion and Future Scope:

1. Implementing AI algorithms to analyze real-time data from sensors for predictive maintenance and optimization of cold storage operations.
2. Utilizing blockchain to create immutable records of temperature and humidity data, enhancing transparency and traceability throughout the cold storage supply chain.

References:

." IoT based Smart Fridge Application " K.Srinivasa Rao(1), M. Bhanu Sridhar (2) , L.Pavani (3), Gayatri Vidya Parishad College of Engineering for Women, International Journal of Engineering Research and Technology(1)(2), Vol.10 Issue 12, December-2021.